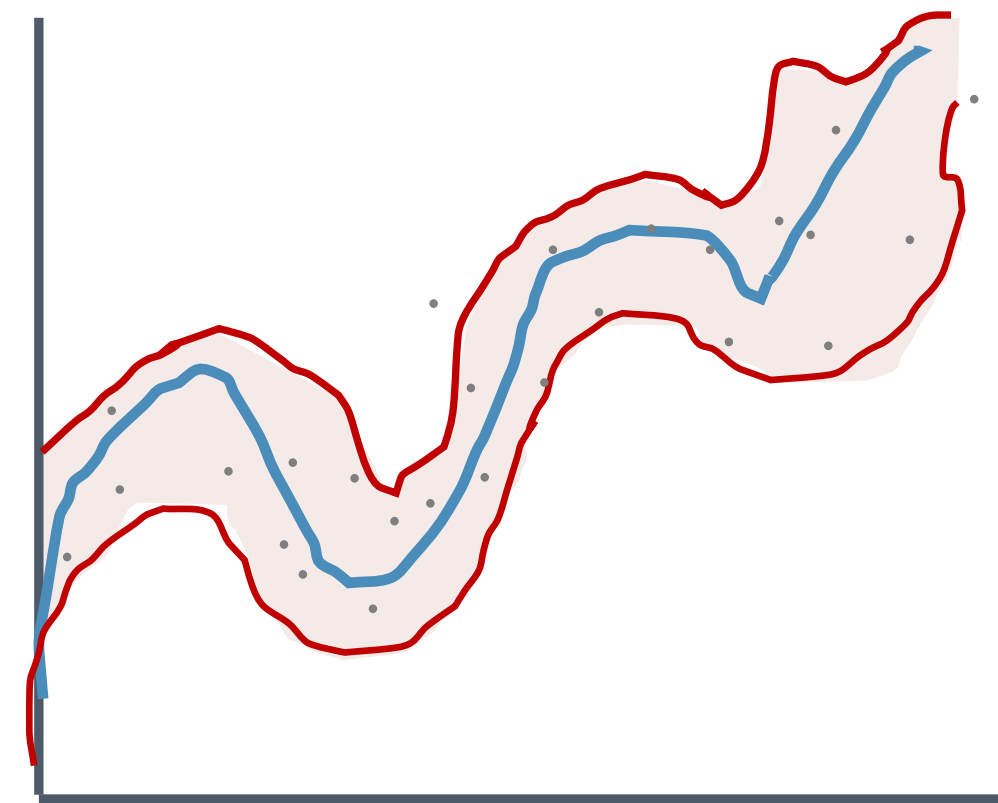


1. Introduction

reliable useful in both stable + changing

Problem

- Conformal prediction (CP) assumes exchangeability.
- Current online CP relies on a single model, which can produce wide (inefficient) intervals.



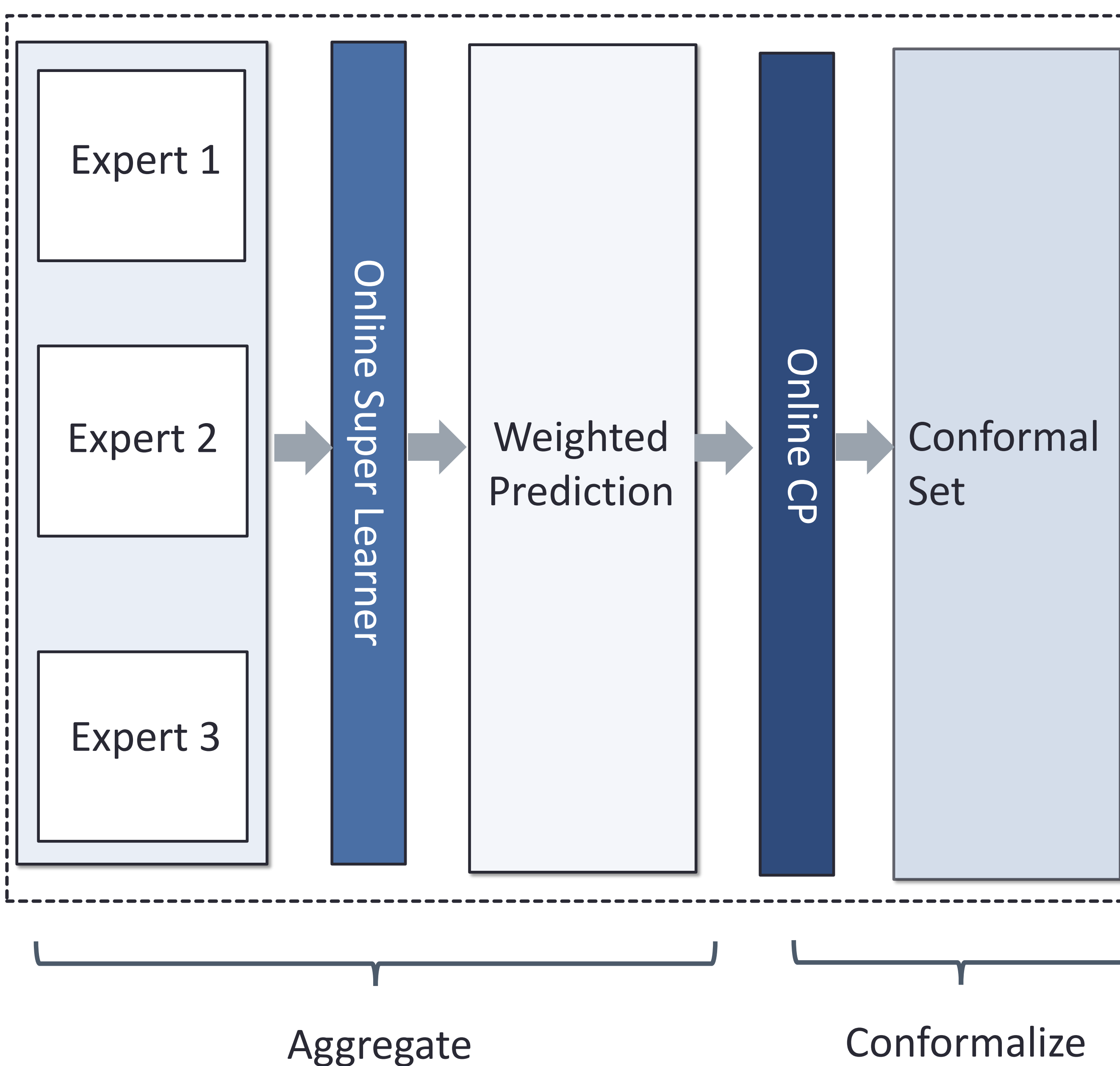
Goal

- Valid and efficient uncertainty quantification method for sequential data.

Contribution

- Aggregate-then-conformalize** approach achieves more efficient (narrower) intervals under both online and i.i.d. settings, outperforming single-model conformal prediction and majority-voting based conformalize-then-aggregate approach (COMA).

(a) One-step online workflow: “aggregate then conformalize”.



2. Methodology

Notation (at time t)

X_t : feature vector

Y_t : observed outcome

$f^{(k)}(X_t)$: prediction from base learner k

$\phi_t^{(k)}$: ensemble weight for learner k

α : target miscoverage level

$$(X_t, Y_t) \in \mathcal{X} \times \mathcal{Y}$$

Online Super Learner

$$\hat{Y}_t^{SL} = \sum_{k=1}^K \phi_{t-1}^{(k)} \hat{f}_{t-1}^{(k)}(X_t)$$

- Initialization:** out-of-fold predictions, $\mathbf{Z}_{\tilde{t}} \in \mathbb{R}^{\tilde{t} \times K}$, on the training set, $\{(X_t, Y_t)\}_{t=1}^{\tilde{t}}$. Weights computed via Non-Negative Least Squares (NNLS)
- Online Update:** for $t = \tilde{t} + 1, \dots, T$, weights obtained via NNLS, $\phi_t = \arg \min_{\phi \in \Delta^{K-1}} \|\mathbf{Y}_t - \mathbf{Z}_t \phi\|_2^2$

$$\phi \in \mathbb{R}^K : \phi_k \geq 0, \sum_{k=1}^K \phi_k = 1$$

Online Conformal Prediction

$$C_t(X_t) = \{y \in \mathcal{Y} : |y - \hat{Y}_t^{SL}| \leq q_t\}$$

- Adaptive Quantile:** q_t targets α . Specifically, $q_{t+1} = q_t + \eta_t (\mathbf{1}\{Y_t \notin C_t(X_t)\} - \alpha)$ for some $\epsilon \in (0, 1/2)$

$$\eta_t \propto t^{-1/2-\epsilon}$$

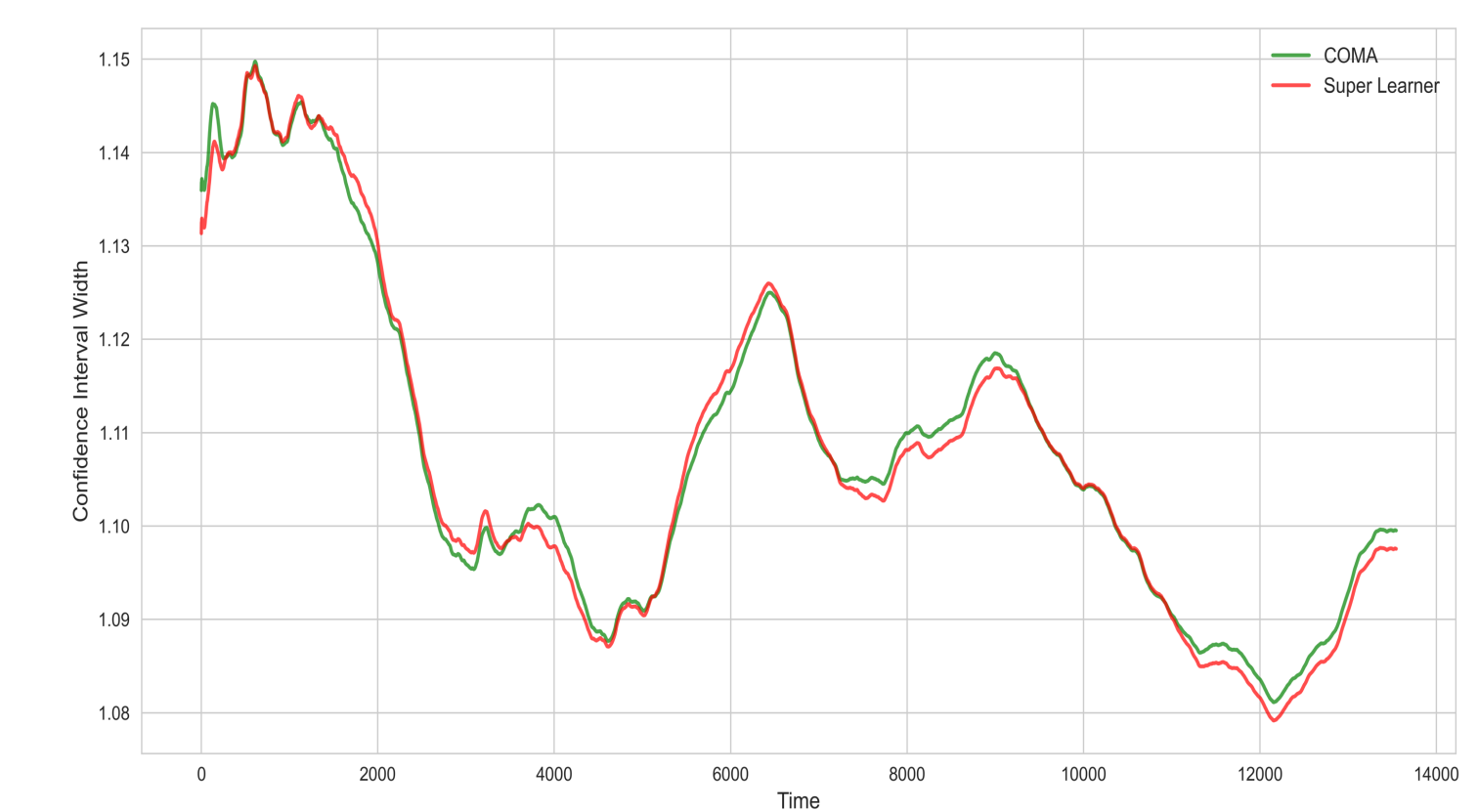
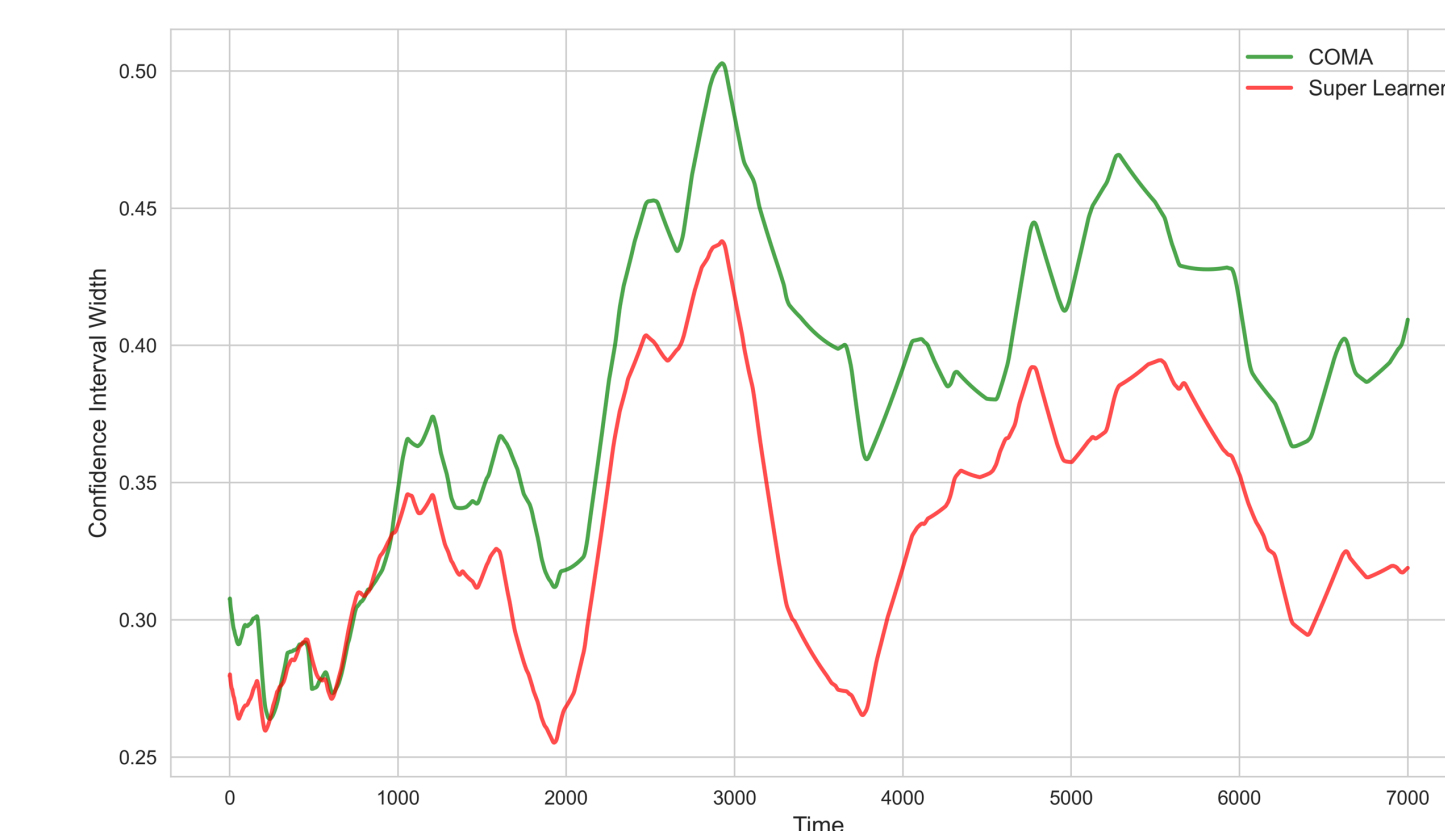
3. Experiment

(a) Mean empirical coverage and CI width on synthetic heteroskedastic dataset.

Method	Mean Coverage	Mean CI Width
GARCH(1,1)	0.9015	0.4702
EGARCH(1,1)	0.9016	0.4710
GARCH(2,2)	0.8966	0.3865
Super Learner	0.9000	0.3311
COMA	0.8948	0.3865

(b) Mean empirical coverage and CI width on i.i.d. Airbnb price dataset.

Method	Mean Coverage	Mean CI Width
Random Forest	0.9000	1.1098
Linear Regression	0.9004	1.4868
Lasso	0.9000	1.6486
Ridge Regression	0.9004	1.4868
Neural Network	0.9006	1.2653
Super Learner	0.9002	1.1087
COMA	0.9000	1.1098



(c) CI width over time (synthetic data)

(d) CI width over time (Airbnb)

Super Learner achieves narrower intervals while maintaining target coverage.

4. Future Work

- To extend ensemble design beyond family-based hyperparameter variation to include structurally different learners (e.g., tree-based or attention models);
- To adapt this method to broader real-world datasets.

5. References

- Gasparin, M. and Ramdas, A. Conformal online model aggregation. *arXiv preprint arXiv:2403.15527*, 2024.
- Angelopoulos, A. N., Barber, R., and Bates, S. Online conformal prediction with decaying step sizes. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 1616–1630. PMLR, 21–27 Jul 2024.